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In [ ]:

In [2]:

```python
import pandas as pd
import numpy as np
import random
from datetime import datetime, timedelta
import os

# Makes sure we get same data every time
np.random.seed(42)
random.seed(42)

NUM_ROWS = 20000

# Things to pick from randomly — like real GL data
gl_accounts    = ['1001-Cash','1200-AR','2001-AP','3001-Equity',
                  '4001-Revenue','5001-COGS','6001-OpEx','7001-Tax']
descriptions   = ['Vendor Payment','Customer Receipt','Payroll','Accrual',
                  'Intercompany Transfer','Adjustment','Reversal','Depreciation']
users          = ['u.sharma','a.mehta','r.das','p.iyer','admin','s.roy','m.khan']
business_units = ['Finance','Operations','HR','IT','Sales']
currencies     = ['INR','USD','CAD']
entry_types    = ['Auto','Manual']

start_date = datetime(2023, 1, 1)
```

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```python
business_units = ['Finance','Operations','HR','IT','Sales']
currencies     = ['INR','USD','CAD']
entry_types    = ['Auto','Manual']

start_date = datetime(2023, 1, 1)

# ── BUILD BASE DATA ──────────────────────────────────────────
data = {
    'JournalID'    : [f'JE{str(i).zfill(6)}' for i in range(1, NUM_ROWS+1)],
    'EntryDate'    : [start_date + timedelta(days=random.randint(0,364),
                      hours=random.randint(0,23), minutes=random.randint(0,59))
                      for _ in range(NUM_ROWS)],
    'GLAccount'    : [random.choice(gl_accounts)    for _ in range(NUM_ROWS)],
    'Description'  : [random.choice(descriptions)   for _ in range(NUM_ROWS)],
    'Amount'       : [round(random.uniform(100,100000),2) for _ in range(NUM_ROWS)],
    'Currency'     : [random.choice(currencies)     for _ in range(NUM_ROWS)],
    'UserID'       : [random.choice(users)          for _ in range(NUM_ROWS)],
    'EntryType'    : [random.choice(entry_types)    for _ in range(NUM_ROWS)],
    'BusinessUnit' : [random.choice(business_units) for _ in range(NUM_ROWS)],
}

df = pd.DataFrame(data)
df['EffectiveDate'] = df['EntryDate']

print(f"Base data created: {len(df)} rows")
```

```
Base data created: 20000 rows
```

In [4]:

```python
# ── INJECT ANOMALIES ──────────────────────────────────────────
```

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```python
print(f"Base data created: {len(df)} rows")
```

```
Base data created: 20000 rows
```

In [4]:

```python
# ── INJECT ANOMALIES ──────────────────────────────────
# These are the "bad" entries our Alteryx flags will catch later

# 1. ROUND NUMBERS (suspicious amounts like 50,000 exactly)
round_idx = random.sample(range(NUM_ROWS), 300)
for i in round_idx:
    df.loc[i,'Amount'] = random.choice([5000,10000,25000,50000,100000])

# 2. WEEKEND ENTRIES (posted on Saturday or Sunday)
used = set(round_idx)
weekend_idx = random.sample(list(set(range(NUM_ROWS))-used), 200)
for i in weekend_idx:
    base = start_date + timedelta(days=random.randint(0,364))
    days_ahead = (5 - base.weekday()) % 7
    sat = base + timedelta(days=days_ahead)
    df.loc[i,'EntryDate'] = sat.replace(hour=random.randint(9,17),
                                        minute=random.randint(0,59))

# 3. AFTER-HOURS ENTRIES (posted between 8PM and 6AM)
used |= set(weekend_idx)
ah_idx = random.sample(list(set(range(NUM_ROWS))-used), 250)
for i in ah_idx:
    base = start_date + timedelta(days=random.randint(0,364))
    hour = random.choice(list(range(20,24))+list(range(0,6)))
    df.loc[i,'EntryDate'] = base.replace(hour=hour,
```

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```python
used |= set(weekend_idx)
ah_idx = random.sample(list(set(range(NUM_ROWS))-used), 250)
for i in ah_idx:
    base = start_date + timedelta(days=random.randint(0,364))
    hour = random.choice(list(range(20,24))+list(range(0,6)))
    df.loc[i,'EntryDate'] = base.replace(hour=hour,
                                         minute=random.randint(0,59))

# 4. DUPLICATE JOURNAL IDs (same ID used twice — a red flag)
used |= set(ah_idx)
dup_idx = random.sample(list(set(range(NUM_ROWS))-used), 100)
for i in dup_idx:
    df.loc[i,'JournalID'] = df.loc[random.choice(
        list(set(range(NUM_ROWS))-{i})),'JournalID']

print(" Anomalies injected:")
print(f"   Round numbers   → {len(round_idx)} rows")
print(f"   Weekend entries → {len(weekend_idx)} rows")
print(f"   After-hours     → {len(ah_idx)} rows")
print(f"   Duplicate IDs   → {len(dup_idx)} rows")
```

```
 Anomalies injected:
   Round numbers   → 300 rows
   Weekend entries → 200 rows
   After-hours     → 250 rows
   Duplicate IDs   → 100 rows
```

In [ ]:

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```
    After-hours     → 250 rows
    Duplicate IDs   → 100 rows
```

In [ ]:

In [6]:

```python
save_path = r'C:\Users\Rim\Downloads\AuditLens\Data\gl_data.csv'

df.to_csv(save_path, index=False)
print(f" File saved successfully!")
print(f"   Location : {save_path}")
print(f"   Rows     : {len(df)}")
print(f"   Columns  : {list(df.columns)}")
```

```
 File saved successfully!
   Location : C:\Users\Rim\Downloads\AuditLens\Data\gl_data.csv
   Rows     : 20000
   Columns  : ['JournalID', 'EntryDate', 'GLAccount', 'Description', 'Amount', 'Currency', 'UserID', 'EntryType', 'BusinessUni
t', 'EffectiveDate']
```

In [7]:

```python
# ── QUICK VERIFICATION ──────────────────────────────
print("=== SHAPE (should be 20000 rows, 10 columns) ===")
print(df.shape)

print("\n=== COLUMNS ===")
print(df.columns.tolist())

print("\n=== FIRST 3 ROWS ===")
print(df.head(3).to_string())
```

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```
t', 'EffectiveDate']
```

In [7]:

```python
# — QUICK VERIFICATION ————————————————————————
print("=== SHAPE (should be 20000 rows, 10 columns) ===")
print(df.shape)

print("\n=== COLUMNS ===")
print(df.columns.tolist())

print("\n=== FIRST 3 ROWS ===")
print(df.head(3).to_string())

print("\n=== ANOMALY SUMMARY ===")
print(f"Round amounts (÷1000):  {(df['Amount'] % 1000 == 0).sum()} rows")
print(f"Manual entries:         {(df['EntryType']=='Manual').sum()} rows")
print(f"Duplicate JournalIDs:   {df['JournalID'].duplicated().sum()} rows")
```

```
--- SHAPE (should be 20000 rows, 10 columns) ---
(20000, 10)

=== COLUMNS ===
['JournalID', 'EntryDate', 'GLAccount', 'Description', 'Amount', 'Currency', 'UserID', 'EntryType', 'BusinessUnit', 'EffectiveD
ate']

--- FIRST 3 ROWS ---
  JournalID           EntryDate  GLAccount           Description    Amount Currency UserID EntryType BusinessUnit       Effect
iveDate
0  JE000001 2023-11-24 03:01:00    2001-AP  Intercompany Transfer  96561.65      CAD  admin      Auto   Operations 2023-11-24 0
3:01:00
```

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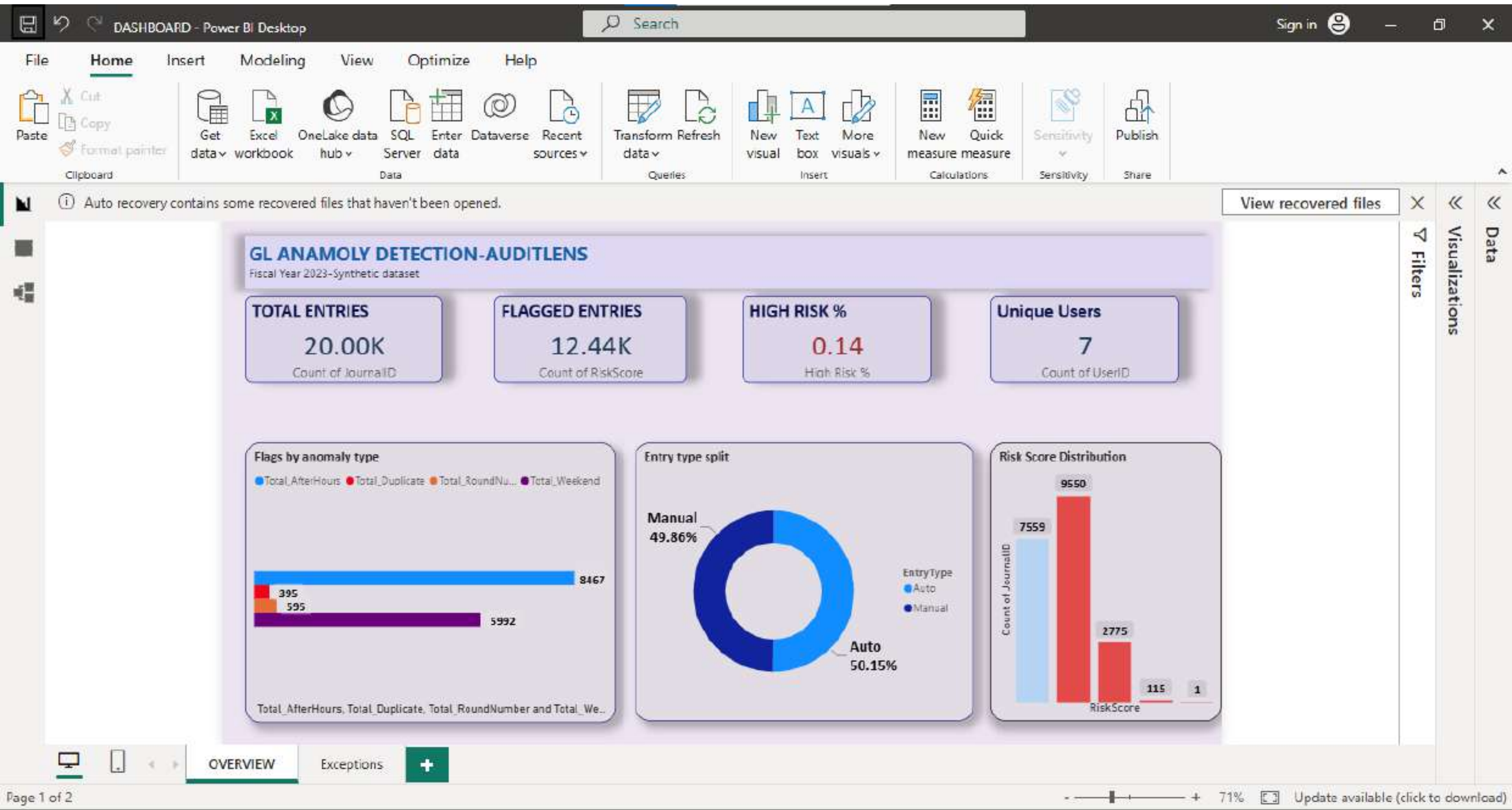
```
=== FIRST 3 ROWS ===
  JournalID           EntryDate  GLAccount           Description    Amount Currency UserID EntryType BusinessUnit        Effect
iveDate
0  JE000001 2023-11-24 03:01:00    2001-AP  Intercompany Transfer  96561.65      CAD  admin      Auto   Operations 2023-11-24 0
3:01:00
1  JE000002 2023-05-21 07:14:00  5001-COGS                Accrual  46491.08      USD  admin      Auto           IT 2023-05-21 0
7:14:00
2  JE000003 2023-03-13 23:06:00  1001-Cash             Adjustment  65096.71      USD  admin      Auto        Sales 2023-03-13 2
3:06:00

=== ANOMALY SUMMARY ===
Round amounts (÷1000):  595 rows
Manual entries:         9971 rows
Duplicate JournalIDs:   198 rows
```

In [8]:

```
print("=== SHAPE (should be 20000 rows, 10 columns) ===")
print(df.shape)
```

```
=== SHAPE (should be 20000 rows, 10 columns) ===
(20000, 10)
```

In [ ]:



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Data
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More visuals
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New measure
Quick measure
Calculations
Sensitivity
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Auto recovery contains some recovered files that haven't been opened.
View recovered files
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Fiscal Year 2023-Synthetic dataset
TOTAL ENTRIES
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Count of JournalID
FLAGGED ENTRIES
12.44K
Count of RiskScore
HIGH RISK %
0.14
High Risk %
Unique Users
7
Count of UserID
Flags by anomaly type
Total_AfterHours
Total_Duplicate
Total_RoundNu...
Total_Weekend
8467
395
595
5992
Total_AfterHours, Total_Duplicate, Total_RoundNumber and Total_We...
Entry type split
Manual
49.86%
Auto
50.15%
EntryType
Auto
Manual
Risk Score Distribution
9550
7559
2775
115
1
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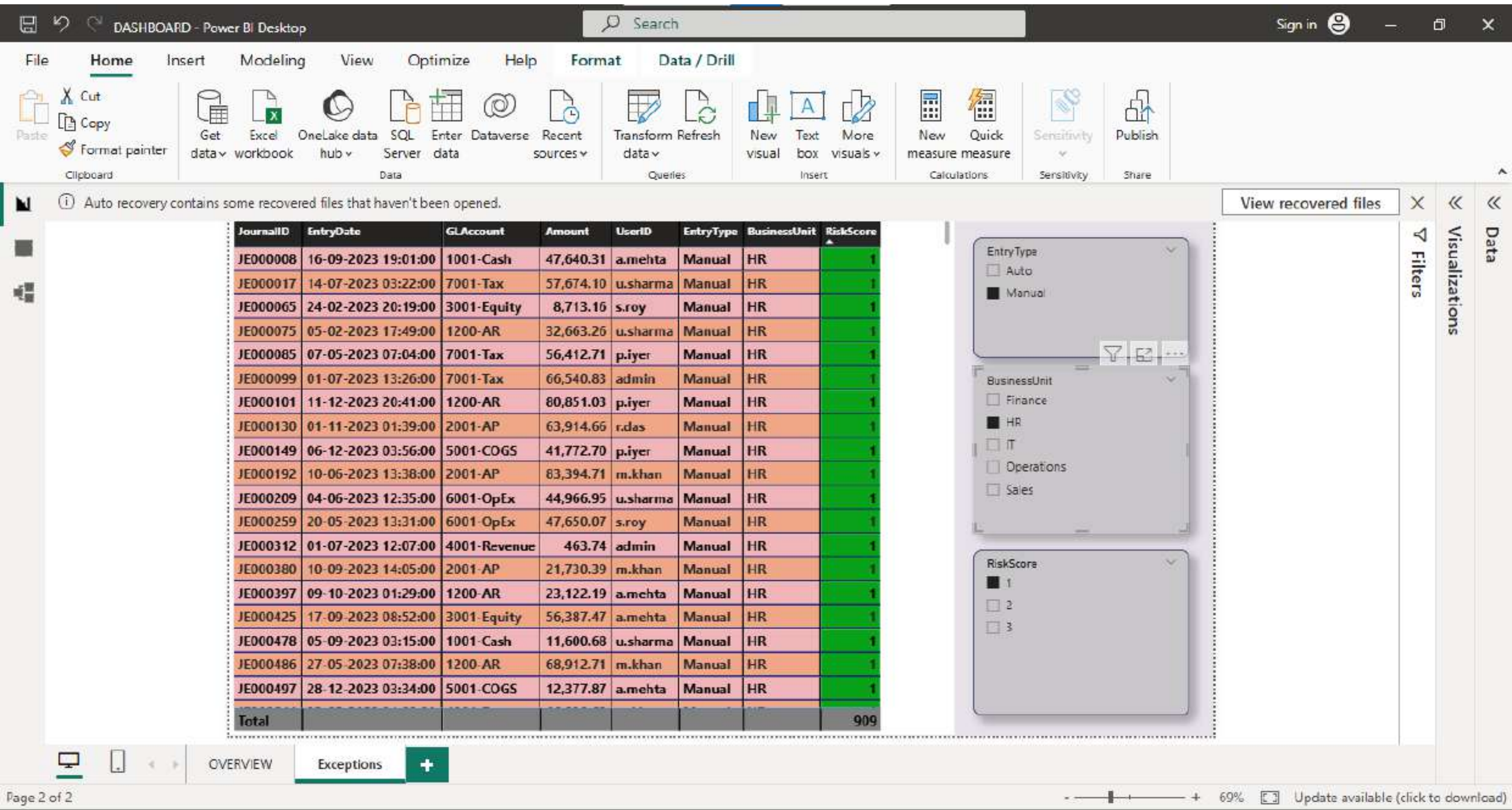
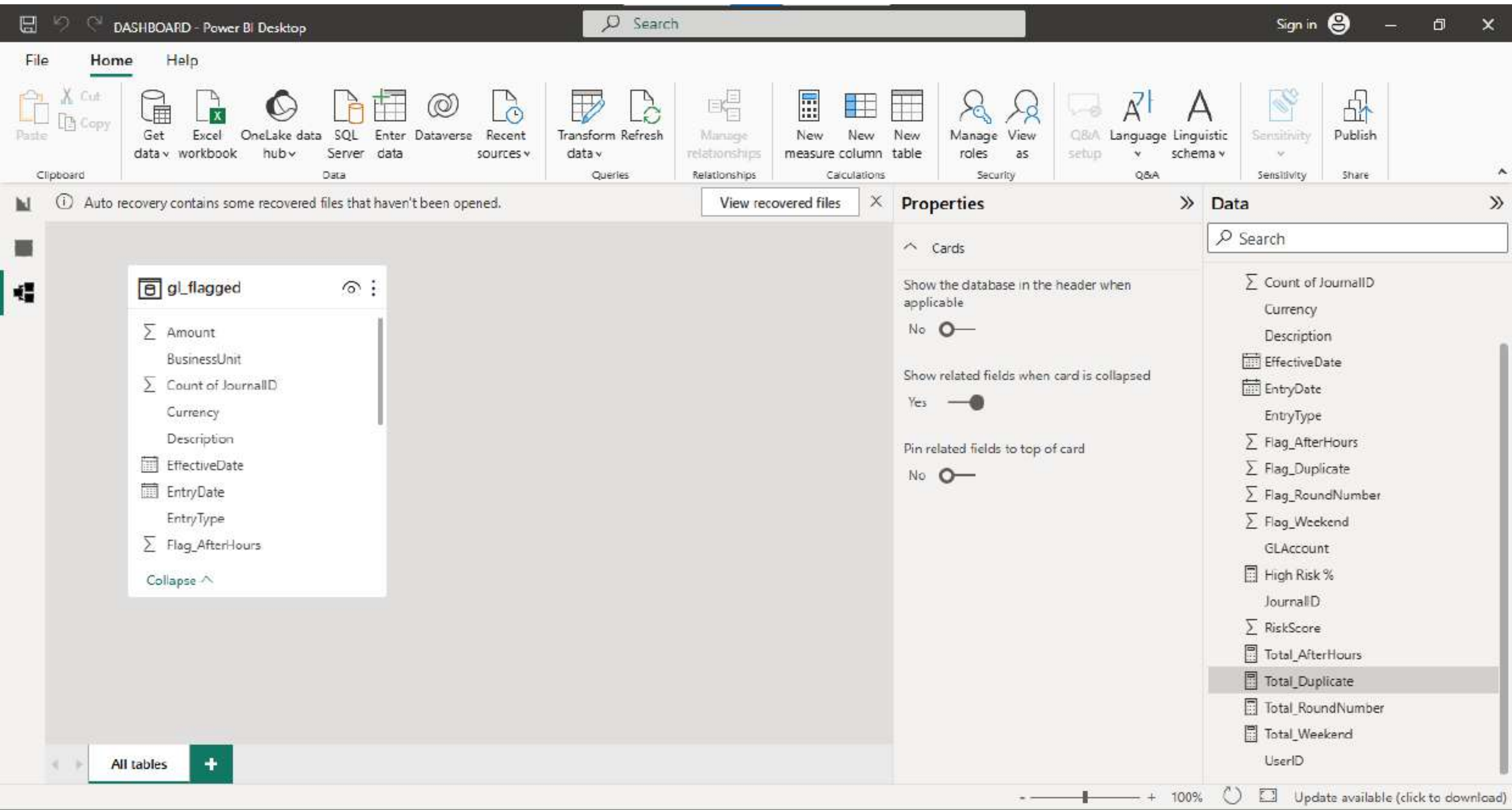


JournalID	EntryDate	GLAccount	Amount	UserID	EntryType	BusinessUnit	RiskScore
JE000008	16-09-2023 19:01:00	1001-Cash	47,640.31	a.mehta	Manual	HR	1
JE000017	14-07-2023 03:22:00	7001-Tax	57,674.10	u.sharma	Manual	HR	1
JE000065	24-02-2023 20:19:00	3001-Equity	8,713.16	s.roy	Manual	HR	1
JE000075	05-02-2023 17:49:00	1200-AR	32,663.26	u.sharma	Manual	HR	1
JE000085	07-05-2023 07:04:00	7001-Tax	56,412.71	p.iyer	Manual	HR	1
JE000099	01-07-2023 13:26:00	7001-Tax	66,540.83	admin	Manual	HR	1
JE000101	11-12-2023 20:41:00	1200-AR	80,851.03	p.iyer	Manual	HR	1
JE000130	01-11-2023 01:39:00	2001-AP	63,914.66	r.das	Manual	HR	1
JE000149	06-12-2023 03:56:00	5001-COGS	41,772.70	p.iyer	Manual	HR	1
JE000192	10-06-2023 13:38:00	2001-AP	83,394.71	m.khan	Manual	HR	1
JE000209	04-06-2023 12:35:00	6001-OpEx	44,966.95	u.sharma	Manual	HR	1
JE000259	20-05-2023 13:31:00	6001-OpEx	47,650.07	s.roy	Manual	HR	1
JE000312	01-07-2023 12:07:00	4001-Revenue	463.74	admin	Manual	HR	1
JE000380	10-09-2023 14:05:00	2001-AP	21,730.39	m.khan	Manual	HR	1
JE000397	09-10-2023 01:29:00	1200-AR	23,122.19	a.mehta	Manual	HR	1
JE000425	17-09-2023 08:52:00	3001-Equity	56,387.47	a.mehta	Manual	HR	1
JE000478	05-09-2023 03:15:00	1001-Cash	11,600.68	u.sharma	Manual	HR	1
JE000486	27-05-2023 07:38:00	1200-AR	68,912.71	m.khan	Manual	HR	1
JE000497	28-12-2023 03:34:00	5001-COGS	12,377.87	a.mehta	Manual	HR	1
Total							909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New column
New table
Calculations
Manage roles
View as
Security
Q&A setup
Language
Linguistic schema
Q&A
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Auto recovery contains some recovered files that haven't been opened.
View recovered files
gl_flagged
Amount
BusinessUnit
Count of JournalID
Currency
Description
EffectiveDate
EntryDate
EntryType
Flag_AfterHours
Collapse
All tables
Properties
Cards
Show the database in the header when applicable
No
Show related fields when card is collapsed
Yes
Pin related fields to top of card
No
Data
Search
Count of JournalID
Currency
Description
EffectiveDate
EntryDate
EntryType
Flag_AfterHours
Flag_Duplicate
Flag_RoundNumber
Flag_Weekend
GLAccount
High Risk %
JournalID
RiskScore
Total_AfterHours
Total_Duplicate
Total_RoundNumber
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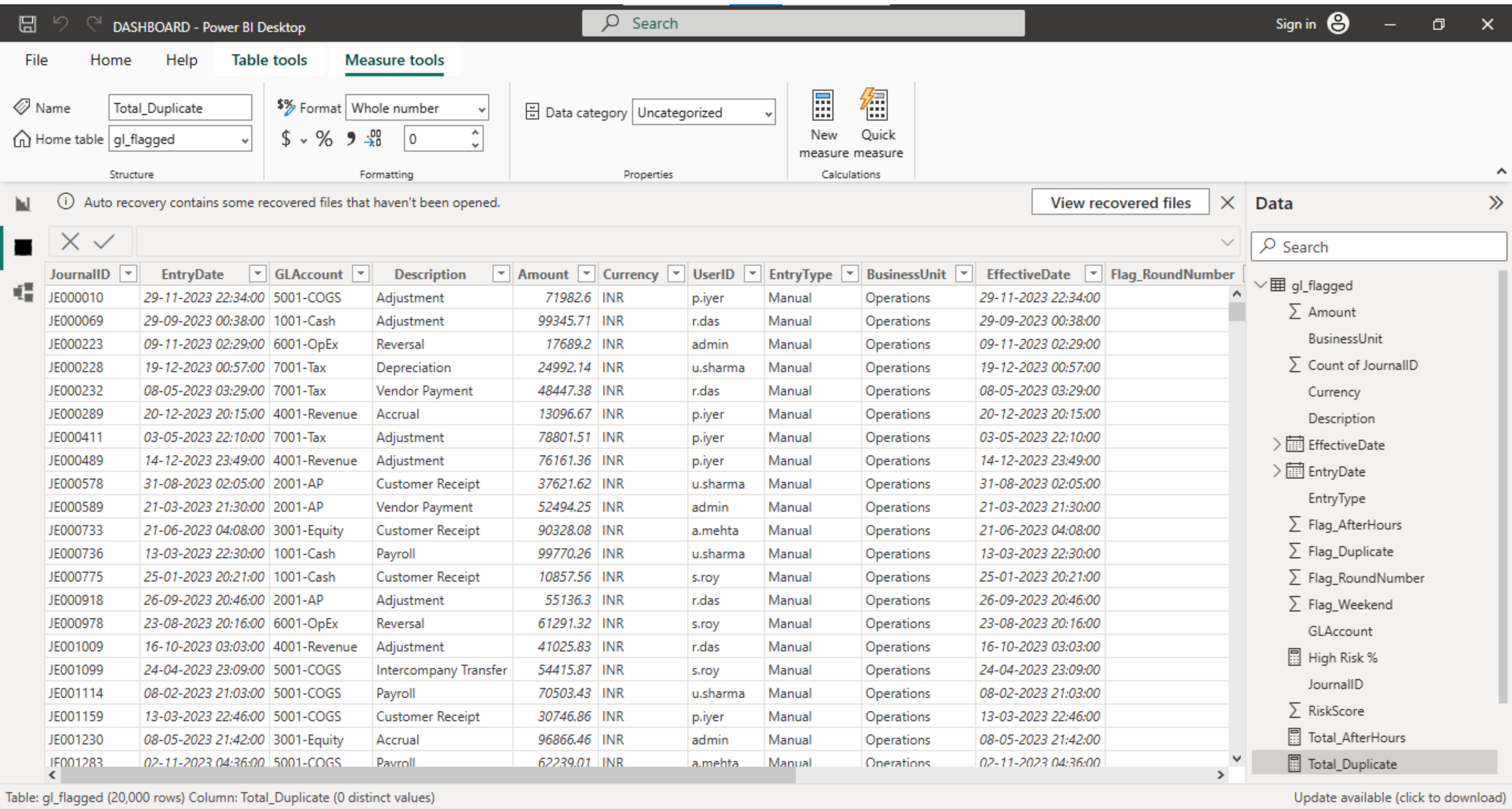

JournalID	EntryDate	GLAccount	Description	Amount	Currency	UserID	EntryType	BusinessUnit	EffectiveDate	Flag_RoundNumber
JE000010	29-11-2023 22:34:00	5001-COGS	Adjustment	71982.6	INR	p.iyer	Manual	Operations	29-11-2023 22:34:00	
JE000069	29-09-2023 00:38:00	1001-Cash	Adjustment	99345.71	INR	r.das	Manual	Operations	29-09-2023 00:38:00	
JE000223	09-11-2023 02:29:00	6001-OpEx	Reversal	17689.2	INR	admin	Manual	Operations	09-11-2023 02:29:00	
JE000228	19-12-2023 00:57:00	7001-Tax	Depreciation	24992.14	INR	u.sharma	Manual	Operations	19-12-2023 00:57:00	
JE000232	08-05-2023 03:29:00	7001-Tax	Vendor Payment	48447.38	INR	r.das	Manual	Operations	08-05-2023 03:29:00	
JE000289	20-12-2023 20:15:00	4001-Revenue	Accrual	13096.67	INR	p.iyer	Manual	Operations	20-12-2023 20:15:00	
JE000411	03-05-2023 22:10:00	7001-Tax	Adjustment	78801.51	INR	p.iyer	Manual	Operations	03-05-2023 22:10:00	
JE000489	14-12-2023 23:49:00	4001-Revenue	Adjustment	76161.36	INR	p.iyer	Manual	Operations	14-12-2023 23:49:00	
JE000578	31-08-2023 02:05:00	2001-AP	Customer Receipt	37621.62	INR	u.sharma	Manual	Operations	31-08-2023 02:05:00	
JE000589	21-03-2023 21:30:00	2001-AP	Vendor Payment	52494.25	INR	admin	Manual	Operations	21-03-2023 21:30:00	
JE000733	21-06-2023 04:08:00	3001-Equity	Customer Receipt	90328.08	INR	a.mehta	Manual	Operations	21-06-2023 04:08:00	
JE000736	13-03-2023 22:30:00	1001-Cash	Payroll	99770.26	INR	u.sharma	Manual	Operations	13-03-2023 22:30:00	
JE000775	25-01-2023 20:21:00	1001-Cash	Customer Receipt	10857.56	INR	s.roy	Manual	Operations	25-01-2023 20:21:00	
JE000918	26-09-2023 20:46:00	2001-AP	Adjustment	55136.3	INR	r.das	Manual	Operations	26-09-2023 20:46:00	
JE000978	23-08-2023 20:16:00	6001-OpEx	Reversal	61291.32	INR	s.roy	Manual	Operations	23-08-2023 20:16:00	
JE001009	16-10-2023 03:03:00	4001-Revenue	Adjustment	41025.83	INR	r.das	Manual	Operations	16-10-2023 03:03:00	
JE001099	24-04-2023 23:09:00	5001-COGS	Intercompany Transfer	54415.87	INR	s.roy	Manual	Operations	24-04-2023 23:09:00	
JE001114	08-02-2023 21:03:00	5001-COGS	Payroll	70503.43	INR	u.sharma	Manual	Operations	08-02-2023 21:03:00	
JE001159	13-03-2023 22:46:00	5001-COGS	Customer Receipt	30746.86	INR	p.iyer	Manual	Operations	13-03-2023 22:46:00	
JE001230	08-05-2023 21:42:00	3001-Equity	Accrual	96866.46	INR	admin	Manual	Operations	08-05-2023 21:42:00	
JE001283	02-11-2023 04:36:00	5001-COGS	Payroll	62239.01	INR	a.mehta	Manual	Operations	02-11-2023 04:36:00	